

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4. NUCLEAR AND PARTICLE PHYSICS

MICHAELMAS TERM 2020

Thursday, 15 October

Opening time: 09.30 am (BST)

You have two hours and thirty minutes to complete the paper and upload your answer file

*Answer **two** questions.*

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors is available at the Physics website.

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) (i) Explain what is meant by a *Nuclear Form Factor*, $F(q)$. (ii) Explain why, for a spherically symmetric charge distribution,

$$F(q) = \frac{C}{q} \int r \rho(r) \sin(qr) dr,$$

where C is a numerical constant that you should determine. [5]

A student suggests that, since protons electrically repel one another, and are free to move within the nucleus, the nuclear charge should be uniformly distributed over a spherical shell on the surface of the nucleus and zero elsewhere.

- (b) Calculate the nuclear form factor associated with (i) the student's charge distribution and (ii) charge distributed uniformly throughout the volume of the nucleus. [9]

(c) Outline the elements of a scattering experiment that could discriminate between these two charge distribution hypotheses, including the beam particle, the beam energy, the geometry of the apparatus, the quantity to be measured, and the expected results in the two cases. You may assume that the radius R of the nucleus is known. [7]

- (d) Discuss the validity of the student's argument about the form of the charge distribution. [4]

2. (a) (i) Describe the assumptions underlying the Fermi theory of beta decay. (ii) Explain why the rates of beta decays are expected to be proportional to the fifth power of the Q -value. [6]

- (b) The table shows some properties of a selection of $J^P = \frac{1}{2}^+$ baryons.

Particle	Mass (MeV/ c^2)	Lifetime (s)	BR ($\rightarrow e^\pm + \text{anything}$)
n		879.4	1
Λ^0		2.63×10^{-10}	8.3×10^{-4}
Σ^0		7×10^{-20}	
Λ_c^+	2286	2.00×10^{-13}	0.045
Λ_b^0	5620	1.45×10^{-12}	0.10

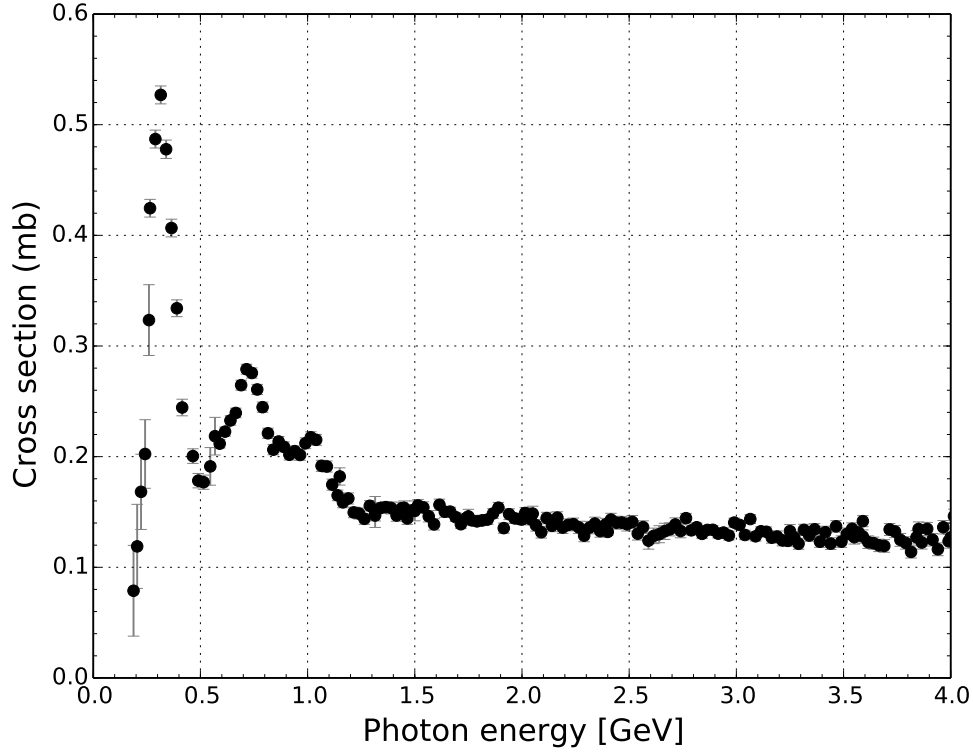
The final column shows the branching ratio to any final state that includes an electron or a positron. The Λ_c^+ is the lightest baryon containing a c quark. The Λ_b^0 is the lightest baryon containing a b quark. (i) Explain how you infer the quark content of each of the five baryons. (ii) Explain why the Σ^0 has the shortest lifetime of the five. [4]

- (c) Draw a leading-order Feynman diagram for the dominant decay to a final state that includes an e^+ or an e^- for (i) the Λ^0 and (ii) the neutron. [3]

- (d) Use the data for the Λ^0 and neutron decays to obtain a value for the Cabibbo angle, stating any assumptions you make. [6]

- (e) Discuss what the data in the table tell us about the mixing between the three generations of quarks. [6]

3. The plot shows the total cross section for photons impinging on protons within a liquid hydrogen target as a function of the energy of the photon.



Consider the lowest-energy resonance visible in the plot.

(a) Use the plot to calculate (i) its mass in units of MeV/c^2 and (ii) its lifetime in seconds. [5]

(b) Given that it is found to have $J^P = \frac{3}{2}^+$, explain how it can be accommodated within the quark model. [5]

(c) Use the maximum cross section to predict the branching ratio of this resonance to $\gamma + p$. [7]

Consider now the data for photon energies larger than about 2 GeV.

(d) Explain the physical interpretation of the lack of structure in this part of the plot. [3]

(e) How would you expect the cross-section to change at these higher photon energies if the liquid hydrogen target was replaced with liquid deuterium? [5]

You may assume that the non-relativistic Breit-Wigner formula for resonances is

$$\sigma = \frac{2J+1}{(2s_1+1)(2s_2+1)} \frac{\pi}{k^2} \frac{\Gamma_i \Gamma_f}{(E-E_0)^2 + \frac{1}{4}\Gamma^2}.$$

4. The semi-empirical mass formula (SEMF) may be written

$$M(A, Z) = Zm_p + (A - Z)m_n - \alpha A + \beta A^{2/3} + \gamma \frac{(A - 2Z)^2}{A} + \epsilon \frac{Z^2}{A^{1/3}} + \frac{\delta}{A^{1/2}}.$$

A fit to the observed masses yields the values:

$$\begin{aligned} \alpha & 15.8 \text{ MeV}/c^2 \\ \beta & 18.3 \text{ MeV}/c^2 \\ \gamma & 23.3 \text{ MeV}/c^2 \\ \epsilon & 0.714 \text{ MeV}/c^2 \\ \delta & \begin{cases} -12 \text{ MeV}/c^2 & \text{for even-even nuclei} \\ 0 & \text{for even-odd nuclei} \\ 12 \text{ MeV}/c^2 & \text{for odd-odd nuclei} \end{cases} \end{aligned}$$

(a) Using the value of ϵ , estimate the value of r_0 in the relation for the nuclear radius $R = r_0 A^{1/3}$. [3]

(b) (i) Derive an expression for the energy released in approximately symmetrical fission of a heavy nucleus. (ii) Use this expression to calculate the energy loss in the fission of ${}^{235}_{92}\text{U}$. (iii) The observed total energy released in the fission of ${}^{235}_{92}\text{U}$ is about 200 MeV. Explain any discrepancy with your calculated value. [6]

(c) Explain why fission can be induced in ${}^{235}_{92}\text{U}$ with thermal neutrons, whereas neutrons of energies $\gtrsim 1$ MeV are required to induce fission in ${}^{238}_{92}\text{U}$. [4]

(d) Explain why the formula implies that for $A \gtrsim 148$ a nucleus will be unstable with respect to α decay. [12]

[The mass of an α particle is $3727.4 \text{ MeV}/c^2$. The electrostatic energy of a ball of radius R with uniformly distributed charge Q is $\frac{3}{5}Q^2/(4\pi\epsilon_0 R)$].